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Cognate facilitation in sentence context – translation production by interpreting trainees and non-interpreting trilinguals

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Conference interpreters form a special case of language users because the simultaneous interpretation practice requires very specific lexical processing. Word comprehension and production in respective languages is performed under strict time constraints and requires constant activation of the involved languages. The present experiment aimed at shedding more light on the effects of conference interpreting training on word production in the process of translation in sentence context. In the study we tested trilingual interpreting trainees and matched non-interpreting trilinguals. Both groups were required to verbally produce L1 and L2 translation equivalents in response to L3 target words (L2–L3 cognates and non-cognates) presented in two context constraints – high sentence context constraint and low sentence context constraint. This enabled us to investigate whether expertise developed during interpreting training in translation between L1 and L2 (extensively practised) is transferable to translation from L3 to L2 (not practised at all). The study provided evidence for cognate facilitation and context effects on naming latencies. However, we found no significant evidence of enhanced semantic processing of interpreting trainees compared to trilingual non-interpreters.

Keywords: cognate facilitation; sentence context; mental lexicon; trilinguals; interpreting; multilingualism

Introduction

One of the central questions in psycholinguistic experimentation with bilingual speakers and even more so with multilingual ones pertains to the nature of lexical processing in their mental lexicons. For years now researchers have been trying to resolve an ongoing debate between two opposing hypotheses i.e. language selective versus language non-selective access to the bilingual/multilingual mental lexicon. Until now, there has been abundant evidence in favour of the language non-selective access hypothesis coming from a number of empirical studies conducted within various paradigms, e.g. in picture naming (e.g. Costa, Santesteban, & Caño, 2005; Hoshino & Kroll, 2008); in eye-tracking (e.g. Bartolotti & Marian, 2012; Ju & Luce, 2004); in word recognition and electrophysiological experimentation (e.g. Hoshino & Thierry, 2012; Peeters, Dijkstra, & Grainger, 2013). Studies on cognate processing and word processing in sentence context can offer further insight into the nature of multilingual mental lexicons.

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The study presented in this article aims to shed more light on the nature of cognate processing in sentence context by trilinguals. We have included conference interpreting trainees in the study to reveal any effect of interpreting training on their lexical processing. Below, we first present an overview of studies focusing on the processing of cognates and sentence context effects on lexical processing both in bilinguals and multilinguals. Their results are reviewed in the context of two models of bilingual lexicon, the Revised Hierarchical Model (Kroll & Stewart, 1994) and the BIA+ model (Dijkstra & van Heuven, 2002). Later, we focus on the studies of lexical processing by interpreters and group differences between professional interpreters, trainees and non-interpreting controls. The actual experiment involving translation of L3–L2 cognates and non-cognates from L3 into L2 and L1 in manipulated context conditions brings interesting data discussed in relation to the two above mentioned models.

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Processing of cognates

One of the key findings interpreted in favour of the language non-selective access is the *cognate facilitation effect*. Cognates are words which share both form and meaning across languages (e.g. the word BUTTER is a cognate for English and German). The typical finding in psycholinguistic experiments investigating reaction times (RTs) to cognates is that they take shorter time to respond to in comparison to language-specific control words. This discrepancy between RTs obtained for cognates and non-cognates is taken to indicate that during lexical access lexicons of all languages known to an individual are activated. The cognate facilitation effect has been found in a number of studies carried out with bilingual speakers (Antón-Méndez & Gollan, 2010; Costa, Caramazza, & Sebastian-Galles, 2000; Davis et al., 2010; De Groot, Dannenburg, & Van Hell, 1994; Dijkstra, Grainger, & van Heuven, 1999; Dijkstra, Miwa, Brummelhuis, Sappelli, & Baayen, 2010; Dijkstra, Timmermans, & Schriefers, 2000; Gollan, Forster, & Frost, 1997; Lemhoefer, Dijkstra, & Michel, 2004; Lotto & De Groot, 1998; Sáchez-Casas, García-Albea, & Davis, 1992; Van Hell & De Groot, 1998; Van Hell & Dijkstra, 2002).

Experiments testing cognate facilitation in trilinguals are scarce (Dijkstra & Van Hell, 2003; Lemhoefer et al., 2004; Poarch & Van Hell, 2012; Szubko-Sitarek, 2011; Tymczyńska, 2011; Van Hell & Dijkstra, 2002). A study by Van Hell and Dijkstra (2002) involved three experiments and two groups of participants. The first group comprised Dutch-English-French trilinguals with a relatively low L3 proficiency and the second group consisted of Dutch-English-French trilinguals whose fluency in French was higher than in the first group and comparable to that in English. Both groups performed lexical decision tasks (LDTs; when participants need to decide if a given string of letters is a word or not) on Dutch words, and, additionally, the first group performed a word association test (i.e. they were asked to produce single word associations in response to L1 words presented on the computer screen). Materials used for the experiments were L1 words which were either Dutch-English cognates, or Dutch-French cognates or non-cognates. All tasks were run in an exclusively L1 context, i.e. participants had not been informed of the multilingual nature of stimuli used in the experiments and no explicit reference to the participants' foreign languages was made in the course of the experimentation. Van Hell and Dijkstra observed a significant cognate facilitation effect for Dutch-English cognates in both groups. The cognate facilitation effect for Dutch-French cognates was recorded only in the second group which had a higher fluency in French. According to Van Hell and Dijkstra (2002) these results lend further support to language non-selective access hypothesis. They claim that the fact that significant cognate

facilitation was observed in an exclusively monolingual context seems to contradict the strict version of the Language Mode Hypothesis (Grosjean, 1998, 2001), which posits that if bilinguals/multilinguals are in a monolingual environment, they are able to deactivate their non-target languages (Van Hell & Dijkstra, 2002, p. 787).

Similar conclusions were drawn from another study conducted with trilingual speakers. Lemhoefer et al. (2004) tested a group of Dutch-English-German trilinguals who performed a LDT in their L3. The participants were asked to decide if the presented string of letters formed a German word or not. The critical stimuli used in the study were cognates overlapping in Dutch and German with no cognate relation to English, as well as cognates overlapping across the three languages. RTs recorded for Dutch-German cognates were significantly shorter than for control German words and significantly longer than RTs for Dutch-English-German ones. This cumulative cognate facilitation effect is even more interesting in view of the fact that the cognate facilitation effect observed for Dutch-English-German cognates was not influenced by whether the participants read an English text and performed an item recognition test before the LDT task or not. According to Lemhoefer et al., the lack of context effect runs counter to the predictions of the Language Mode Hypothesis put forward by Grosjean (1998, 2001), which assumes that the context of experimental situation should affect activation levels of languages in the participants' minds. All in all, Lemhoefer et al. (2004) interpreted the results of their experiments in favour of the language non-selective access in trilingual processing.

The cognate facilitation effect has been accounted for by a number of models of bilingual lexical processing, the most influential being the Revised Hierarchical Model (RHM) (Kroll & Stewart, 1994) and BIA+ (Bilingual Interactive Activation Plus) model (Dijkstra & van Heuven, 2002). Even though these two models have been extended to multilingualism (for details see Tymczyńska (2012) and Dijkstra (2003), respectively), accounts of cognate facilitation within the models will be presented within the bilingual context (i.e. in the form they were originally put forward).

The cognate facilitation effect has been accommodated by RHM for L2 processing in the following manner. Due to direct L1–L2 lexical links, the meaning of an L2 word is accessed more quickly when there is an overlap in lexical representations (word forms) of L1–L2 translation equivalents (as in cognates) as compared to the lack of such an overlap (as in non-cognates). According to Dijkstra et al. (2010), cognates differ from non-cognates within the BIA+ model in that the former share more orthographic, semantic and/or phonological features with their translation equivalents than the latter. The cognate facilitation effect is accommodated by the model in the following manner (Dijkstra et al., 2010, p. 286):

> the cognate facilitation effect in reading might in fact be an orthographic–semantic priming effect: overlapping orthographic and semantic representations of both languages become active upon the presentation of one of the readings of the cognate, leading to a facilitated recognition of cognates relative to non-cognates.

Consequently, the cognate facilitation effect is sensitive to frequency as well as cross-linguistic similarity (both orthographic and phonological) of translation equivalents. BIA+ was supported in cognate studies conducted by e.g. Dijkstra et al. (2010) and Voga and Grainger (2007). Even though Dijkstra et al. (2010) explained the mechanics of cognate facilitation within BIA+ for bilinguals, there is every reason to assume that a similar pattern of processing will also work for multilinguals within the Multilingual

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Interactive Activation model (MIA) which is the multilingual extension of BIA (Dijkstra, 2003).

Sentence context effects

It needs to be stressed here that the vast majority of the bilingual/multilingual research tested cognates and non-cognates by presenting them as single words. Only in a relatively small number of studies did the researchers investigate cognates in a sentence context. Given the fact that in everyday, i.e. non-laboratory language use, speakers extremely rarely use isolated words, investigating lexical processing in sentence context seems to have more ecological validity than experimentation with single words.

Nevertheless, studies testing cognates in sentence context are still very limited in number and differ in terms of the chosen experimental paradigms, e.g. word naming (Schwartz & Kroll, 2006), LDT (Duyck, Van Assche, Drieghe, & Hartsuiker, 2007; Van Hell & De Groot, 2008), translation (Van Hell & De Groot, 2008; Van Hell, 2005), eye-tracking (Duyck et al., 2007; Libben & Titone, 2009; Van Assche, Drieghe, Duyck, Welvaert, & Hartsuiker, 2011). The results recorded in these studies differ across paradigms and do not always tally with findings from single word experiments.

Results obtained from word naming suggest that the processing of cognates is affected by context only if this context is not semantically constraining (i.e. low context, henceforth LC) as in 'The final word in this sentence is…'; where the sentence final target word cannot be easily predicted. As opposed to 'She took a bite of the fresh green…' which is a semantically constraining context (i.e. high context, henceforth HC) biased towards the final target word (APPLE). Schwartz and Kroll (2006) tested two groups of Spanish-English bilinguals in word naming in L2. The materials included identical and non-identical cognates as well as Spanish-English interlingual homographs. The words were embedded in HC and LC sentences. The sentences were presented using the Rapid Serial Visual Presentation (RSVP, where words appear on the screen one-by-one) and the participants were asked to read out the target word. RTs recorded in the study indicated a robust cognate facilitation effect in the LC condition; however, in the HC condition the effect was eliminated. Schwartz and Kroll (2006) interpreted this finding in favour of the language non-selective access which is restricted when enough semantic information is available to suppress the non-target language representations.

A similar pattern of results was obtained in an LDT. In one of the experiments reported on by Van Hell and De Groot (2008) a group of Dutch-English bilinguals performed an L2 LDT where they were asked to decide if a letter string formed an English word or not. Target words were either (concrete or abstract) cognates or non-cognates embedded in HC, LC sentences or presented in isolation (no context condition, NC). Significant cognate effect was observed only in the LC and NC conditions both for concrete and abstract target words. Van Hell and De Groot (2008) found no significant three-way interaction between concreteness, cognate's status and context. This might indicate that subtle meaning overlap variations between cognates and non-cognates do not influence the pattern of results.

In the same paper Van Hell and De Groot (2008) reported on three experiments involving L1–L2 (forward) and L2–L1 (backwards) translation. Materials and participants were similar to those in the LDT discussed above. Unlike in word naming and LDT, cognate facilitation effect was observed across all contexts and translation directions but in the HC condition the facilitation was heavily reduced.

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Similarly to other methodologies, studies employing eye-tracking methods have also yielded discrepant results as far as cognate processing in context is concerned. Libben and Titone (2009) and Titone, Libben, Mercier, Whitford, and Pivneva (2011) recorded eye movements of French-English bilinguals when they read L2 sentences with cognates, interlingual homographs and matched control words. Early-stage comprehension measures (e.g. first fixation duration, gaze duration, and skipping) as well as late-stage comprehension measures (e.g. go-past time and total reading time) showed significant cognate facilitation for LC sentences. In the HC condition only the early stage comprehension measures indicated cognate facilitation. In contrast, Van Assche et al. (2011), who also investigated cognate processing with the use of eye-tracking, found cognate facilitation effect in both contexts for both early- and late-stage comprehension measures in their study. The most general observation from the aforementioned studies seems to be that only semantically rich context exerts influence on cognate facilitation.

Unfortunately, to the best of our knowledge no studies concerning the influence of context on cognate processing with trilinguals have been conducted yet. Hence the present study aims at filling this gap. However, before a detailed account of our experimentation is provided, the paragraphs to follow will briefly introduce interpreters as a special case of multilinguals since they constituted one of the groups tested in the present study.

Lexical processing by interpreters

Interpreters and interpreting trainees are deemed a special case of bilinguals/multilinguals due to their specific use of languages known to them (Chmiel, 2010). According to Paradis (1994), interpreters are required to complete tasks 'that bilinguals, in order to minimise interference, try to avoid' (p. 319). This specific use of two or more languages, where the language systems for the source language and target language have to be deliberately kept active, makes conference interpreters and trainees an interesting research group that can offer valuable insights into language processing by multilinguals.

A number of studies have compared professional interpreters with trainees and non-interpreters by looking at the performance of all groups in interpreting tasks (Dillinger, 1994; Fabbro, Gran, & Gran, 1991; Jörg, 1997; Padilla, Bajo, Canas, & Padilla, 1995; Riccardi, 1998; Sunnari, 1995) and found consistent superiority of interpreters over trainees as well as trainees over non-interpreters reflected in smoother delivery, prioritising information, effective cognitive capacity management, deeper semantic processing, easier restructuring of syntax and better verb anticipation. Interpreting practice thus seems to accelerate various linguistic tasks, including semantic processing.

Studies involving professional interpreters and non-interpreting bilinguals performing single word translation tasks bring inconclusive results. Christoffels, De Groot, and Kroll (2006) found shorter translation latencies for interpreters performing English-Dutch (L2–L1) and Dutch-English (L1–L2) word translation as compared to bilingual non-interpreting students but no difference between interpreters and English teachers. This suggests that single word translation is a skill that can be boosted not only through extensive interpreting practice but also through concurrent use of both languages in another context, i.e. foreign language teaching. Moreover, Christoffels et al. (2006) found asymmetries according to translation direction for the (non-interpreting) student group only. Translation into L2 was faster than into L1 offering further support for the RHM-based suggestion of asymmetrical lexical links.

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One of the variables manipulated in the study by Christoffels et al. (2006) was cognate status. A general cognate effect was found. In both translation directions the effect was smaller for high frequency words than for low frequency words. This pertained to all experimental groups, i.e. interpreters and non-interpreting bilinguals (both students and English teachers). The authors speculated that the cognate effect might be attenuated for interpreters if they are able to better control the non-target language. However, no interaction between group and cognate status was found, thus confirming that the cognate effect size does not depend on group characteristics.

The issue of language control, language switching, activating and inhibiting the appropriate systems is very crucial when more than one language comes into play in production. According to De Groot and Christoffels (2006, p. 194):

> bilingual language production requires the suppression of the non-target language and the activation of the target language, and (…) the degree of suppression of the non-target language depends on language strength: the stronger the non-target language, the more it needs to be suppressed in order not to interfere with the current target language.

The same authors suggest that translation equivalents (both words and multiword units) are linked via direct memory connections since they tend to co-occur, which forms associations between them. In fact, 'the more often the same two terms (words or longer phrases) co-occur in a translation act, the stronger the memory connection between them will be' (De Groot & Christoffels, 2006, p. 198).

The above suggestions are largely supported by Paradis (1984), who also claims that during production both languages are activated and the unspoken language undergoes inhibition. Paradis (1994) proposed the Activation Threshold Hypothesis, according to which any trace (in our case understood as the word) will have its activation threshold depending on its frequency of activation and time since its last activation (p. 320). In other words, the more often and the more recent the activation of a given word, the lower will be its activation threshold. However, for the successful production of a word its competitors have to be inhibited. According to Paradis (1994), if a given item is activated, its competitors are inhibited, and as the word's activation threshold decreases, the threshold for the competing items increases. The same pertains to the whole language system. The non-selected language system is inhibited 'to prevent interference during production, but not sufficiently to preclude borrowing (…) or comprehension in the other language' (Paradis, 1994, p. 321). Interpreters are thus different from non-interpreting bilinguals because while the latter inhibit the non-selected language to facilitate production and comprehension in the selected one, the former keep both language systems activated with varying activation thresholds. Paradis (1994) even suggests that 'the form of the SL [source language] in the short term memory (STM) seems to inhibit to some extent access to its translation equivalent' (p. 322).

A lot of research in this area both in psycholinguistics and Interpreting Studies has been done with reference to two languages. One of the few studies testing the influence of interpreting experience in the trilingual context was the one reported by Tymczyńska (2012). She tested professional interpreters, conference interpreting students and non-interpreting trilinguals in a single word translation task from L1 to L3. She manipulated the cognate status of target translations, i.e. some L3 target translations were cognates with their L2 equivalents but not L1 stimulus words. A moderate L2–L3 cognate effect was found in all experimental groups in the L1–L3 translation task.

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The present study

The present study was designed to address research questions concerning the effects of interpreting training and sentence context on translation as well as the influence of sentence context on cognate facilitation. Our participants were asked to translate L3 words into L1 and L2, even though this kind of translation is not a part of the interpreting training half of our participants (i.e. interpreting trainees) went through. By choosing L3–L2 and L3–L1 translation for the purposes of our experimentation, we wanted to test whether skills developed during interpreting training (in L1–L2 and L2–L1 translation) such as e.g. accelerated semantic processing are potent enough to boost performance of interpreting trainees in the unpractised translation direction (from L3 to L1 and from L3 to L2).

Admittedly, the task applied does not fully reflect what actually happens when interpreting. However, we had to simplify the experimental task in order to avoid numerous otherwise uncontrolled and confounding variables. The main difference between the experimental task (translation of the final word in a sentence) and conference interpreting is that, in real life, interpreters interpret whole texts and not single words. The sentences used in the study do not form a coherent text and interpreters are only asked to provide translations of the final words to measure their production latencies with satisfactory precision. Such a design guarantees, in our opinion, the control of the variables important for our study and can offer insights about lexical production by interpreters under various context constraints and about how this production is influenced by the interpreting experience. De Bot (2000, p. 80) supports this approach and claims that studying word translation can help us understand the nature of interpreting since language production and perception are lexically-driven processes 'highly dependent on the speed of accessing of words in the lexicon'.

It has to be noted that we use the term translation in line with the psycholinguistic tradition – in Translation Studies, or in Interpreting Studies the task applied in the present study would be called interpretation as it pertains to an oral rendition of the source text into the target text. To be even more precise, since the stimuli are visual and not aural, the task is actually sight translation, i.e. oral rendition of the source text presented visually. Sight translation actually happens when interpreters are asked to interpret excerpts of written documents at a meeting.

Additionally, it is worth mentioning that interpreting is frequently viewed in the Translation Studies tradition as a three-stage activity, e.g. involving comprehension, deverbalisation (perception of sense without the verbal form) and production (cf. Seleskovitch & Lederer, 1995). However, the latter theory has been extensively criticised (Pöchhacker, 2004; Setton, 2003). Also, the present study builds upon abundant psycholinguistic research confirming language non-selective access and cognate effect in linguistic processing. Thus, in line with the psycholinguistic approach, it is assumed here that translation (as used in the form of an experimental task in the present study) involves comprehension and production stages only.

Research questions

In view of the experiments and their findings discussed above, the present study aimed to address the following questions:

(1) Will L3–L2 cognates be translated faster than non-cognates in translation into both target languages (L2 and L1)?

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(2) Does sentence context constraint affect cognate processing in translation?
(3) Do interpreting trainees use context more efficiently than non-interpreters in translation from their L3 to L1 and L2?

Method

A 2 (group: interpreting trainees, non-interpreting trilinguals) by 2 (context constraint: high context constraint, low context constraint) by 2 (target language: L2, L1) by 2 (cognate status: cognate, non-cognate) mixed factorial design was used with group as a between-group variable and context, target language and cognate status as within-subject variables.

Participants

Two groups of participants took part in the present study. The group of interpreting trainees (henceforth INT) included 23 interpreting students (6 male, 17 female) with the mean age of 22.1. All of them were students of the Institute of Applied Linguistics at Adam Mickiewicz University. They were enrolled in the final year of the BA programme with Applied Linguistics as their major and conference interpreting as an additional specialisation. Their exposure to conference interpreter training was at least 150 contact hours (note-taking, German-Polish and Polish-German consecutive interpreting, German-Polish and Polish-German simultaneous interpreting) over the course of one year. The group of non-interpreting trilinguals (henceforth NON-INT) comprised 20 students enrolled in the penultimate and the final years of the BA programme (3 male, 17 female; mean age 21) at the Institute of German Studies at Adam Mickiewicz University.

All participants from both groups had Polish as L1, German as L2 and English as L3. Their German competence was at the C2 level within the Common European Framework of Reference for Languages (CEFR). Such a high level of German competence was a prerequisite for university programmes the participants were enrolled in. Their English was not lower than the CEFR level B2 since this was the level of the English classes they all attended. Furthermore, all students scored at least 80% at the English test administered before the experiment (suggesting that their language proficiency in English was at least at the upper-intermediate level). Other languages known to the participants included: Spanish, Russian, French, Swedish, Italian, Finnish but the participants themselves considered them their L4s (acquired last and at the lowest level of proficiency of all languages known to them at the time of the experiment).

Materials

Forty-four English concrete nouns matched for frequency and length were selected for the experiment (these were taken partially from Kujałowicz, Chmiel, Rataj, & Bartłomiejczyk, 2008). Word frequencies were taken from the CELEX lexical database (Baayen, Piepenbrock, & Gulikers, 1995). Half of these words were L3–L2 cognates (e.g. UNCLE-ONKEL) and half of them were non-cognates (e.g. PEAR-BIRNE). The cognate status of the words was assessed by a group of 20 trilinguals taken from a population similar to the participants of the present study. They used a seven-point Likert scale to decide on the similarity of translation equivalents (1-very low similarity, 7-very high similarity). The trilinguals were asked to use both spelling and pronunciation as similarity criteria. The mean score was 6.3 out of 7.

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The experimental stimuli appeared as final words in English sentences (taken partially from Kujałowicz et al., 2008) with two types of context constraint: high – HC (example: She loves pepperoni pizza with double CHEESE.) and low – LC (example: Stir in some broccoli, tomatoes and CHEESE.). Each word appeared in HC and LC but no participant saw the same word twice. Both the sentences and the target words were presented in black font against a white background.

The type of context constraint in the sentence was established through a probability cloze test performed by 83 Polish-English bilinguals. They were presented with sentences without the final words and were asked to complete the final words so that the sentence would make sense. If at least 70% of the respondents completed a given sentence with the same word (also happening to be one of the experimental stimuli) that sentence was considered a high context constraint. If less than 30% of the respondents completed a given sentence with the same word that sentence was considered a low context constraint. The sentences were also judged by a native speaker of English for their accuracy and naturalness. Examples of HC and LC sentences with cognates and non-cognates used in the study are given in Table 1.

Apparatus and procedure

The experiment was run on a Fujitsu-Siemens laptop in a normally lit room or a simultaneous interpreting booth in a classroom. Each participant was tested individually and asked to produce a verbal response (translation equivalent) into a microphone. Reaction times were recorded by means of a voicekey (Serial Response Box). The stimuli were presented in black on a white screen in E-Prime 2.0 (Schneider, Eschman, & Zuccolotto, 2002).

The procedure within trial was as follows. First, the participants saw a fixation point for 500 ms and then they saw a sentence without the final target word. The presentation of the sentence was self-paced, i.e. the participants were asked to read the sentence silently and carefully and press a spacebar immediately after having finished reading it. Then another fixation point appeared for 500 ms, followed by the target word to be translated orally into the respective language as quickly and as accurately as possible. The word was displayed until the participant's verbal response was registered by the voicekey. If no response was given for 10,000 ms, the target word disappeared and the next trial began.

The presentation of sentences was blocked by target language and counter-balanced across participants. Each block contained 22 sentences (11 HC and 11 LC pseudo-randomised sentences) where half of the target words were cognates and half were

Table 1. Example of HC and LC sentences with cognates and non-cognates used in the study.

	High context	Low context
Cognate	On sunny days they picnic sitting on the green GRASS.	In the background she painted the GRASS.
	Nice flowers and vegetables grow in her GARDEN.	The children love playing in the GARDEN.
Non-cognate	Only few survivors reached the island's SHORE.	There were several trucks near the SHORE.
	She loves pepperoni pizza with double CHEESE.	Stir in some broccoli, tomatoes and CHEESE.

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non-cognates. Each experimental block was preceded by a practice block with 10 sentences similar to those used in the experiment proper.

The participants were asked to complete a language history questionnaire. Then task instructions were provided in L3 at the beginning of the experiment. The participants were asked to provide an accurate translation of the target word as quickly and accurately as possible. All in all, each participant performed two trial blocks (translation into L2 and translation into L1) preceded by practice blocks. Each trial block was followed by a recognition task to check if the participants read the sentences carefully enough to process them semantically. The whole experimental procedure took no more than 25 minutes.

Results and discussion

For each participant, mean RTs were calculated for each of the six conditions – context (HC vs. LC) by target language (L2 vs. L1) by cognate status (cognate vs. non-cognate). The data were trimmed as follows: observations following failed voicekey response or wrong verbal response (wrong translation, gap fillers that activated the voicekey before the actual response, etc.) were eliminated. Observations longer than 2.0 SD above the participant's mean and longer than 2000 ms were also eliminated as outliers. It turned out that accuracy for L3–L2 translation was below 75% so it was decided to remove this condition from the present analysis. Hence, for further analysis, RTs recorded for each of the four conditions – context (HC vs. LC) by cognate status (cognate vs. non-cognate) were considered.

We performed a mixed design ANOVA just for the L3–L1 translation direction on the mean subject RTs with context and cognate status as within-subject variables and group as a between-subjects variable. The mean RTs as well as error rates are presented in Table 2.

Even though numerically patterns of cognate facilitation seem different, we found no significant main group effect, $F(1,41) = .20$, $p > .05$, which means that the groups did not differ above the chance level. The analysis also yielded a significant context effect, $F(1,41) = 7.90$, $p < .05$, but the interaction between group and context failed to reach significance, $F(1,41) = 0.37$, $p > .05$. This means that translations were produced at different speeds in HC and in LC but the pattern of these mean RTs did not differ between groups. The main cognate effect turned out to be significant, so generally cognates were translated faster than non-cognates ($F(1,41) = 44.77$, $p < .000$). However, the interaction between cognate status and context turned out to be insignificant ($F(1,41) = 0.56$, $p > .05$.), showing that cognate facilitation was not affected by context. A priori planned

Table 2. Mean RTs (in ms), SDs (in parentheses) and error rates (in %) in each condition in the two groups.

Groups	Conditions		Mean (SD)	ER
INT	HC	Cognates	860 (212)	2.8
		Non-cognates	902 (211)	8.4
	LC	Cognates	1017 (212)	5.6
		Non-cognates	1091 (202)	19.6
Non-INT	HC	Cognates	810 (184)	1.4
		Non-cognates	900 (142)	9.8
	LC	Cognates	1034 (201)	7.7
		Non-cognates	1044 (159)	18.9

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simple effect analyses in the form of paired-samples t-tests (with Bonferroni correction applied, resulting in a significance level set at $p < .025$) revealed no significant cognate facilitation effects in neither of the context conditions ($t(42) = -2.26$, $p > .25$ for HC condition and $t(42) = -2.09$, $p > .25$ for LC conditions, respectively). The significant context effects corroborate the findings by Van Hell and De Groot (2008), Van Hell (2005) and Schwartz and Kroll (2006) suggesting that a highly predictable sentence context facilitates translation production. The main cognate effect was also significant, offering further support to Dijkstra and Van Hell (2003), Lemhoefer et al. (2004) and Van Hell and Dijkstra (2002). Detailed explanation of the fact that L3–L2 cognate facilitation was recorded in L3–L1 translation is given in the discussion of results of the redefined study below.

As mentioned above, low accuracy for L3–L2 translation resulted in the elimination of the data and the target language variable from the analysis. As far as interpreting trainees are concerned such a low accuracy rate may suggest that these participants had problems with this translation direction since they had rarely interpreted from L3 to L2 since interpreting from L3 is generally practised only into L1 and not L2. The interpreter training that our trainees were exposed to only included L1–L2 and L2–L1 directions.

Another explanation might be offered by the Activation Threshold Model by Paradis (1994). According to the model translation equivalents in the interpreter's working languages are generally activated in interpreting (for comprehension or production purposes, depending on the source and target languages). Inhibition is the mechanism that should successfully prevent interference from the source language in the production in the target language. In the present study, it might have been the case that L3 stimuli inhibited otherwise activated L2 equivalents to avoid potential interference, which resulted in offsetting cognate facilitation.

The low accuracy of the non-interpreting trilinguals in interpreting into L2 could be explained by their insufficient L3 competence (B2 level), but a more plausible explanation is (which may also be true for the interpreting group) that the experimental stimuli in L3 inhibited L2 translation equivalents.

We found no group effect and no influence of interpreting training on word production by trilinguals in any condition which suggests that the training period or its intensity was insufficient to influence the processes involved in lexical processing of this type. The lack of differences between interpreters and non-interpreters might stem from the nature of interpreter training to which our participants were exposed before the study. It involved only L2–L1 and L1–L2 translation and skills thus developed might not be transferable to L3–L2 or L3–L1 translation. This issue calls for further research on the nature language processing by interpreters and non-interpreters.

The study redefined

As mentioned, the effect of target language could not be analysed as it had been planned due to low accuracy (<75%) in the L3–L2 translation part of the study. However, we redefined the experiment in order to be able to analyse the translation direction from a slightly different perspective. Namely, those participants of the original experiment whose accuracy in both translation directions was above 75% were selected and their naming latencies were re-analysed. Responses of other participants were excluded from the new analysis. In the paragraphs to follow, a detailed report of the redefined experiment will be provided.

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Method

A 2 (cognate status: cognate, non-cognate) by 2 (context: high constraint, low constraint) by 2 (target language: L1, L2) factorial design was used.

Participants

Twenty participants (12 interpreting students and 8 non-interpreting trilinguals) whose accuracy in translation into both target languages was above 75% were selected out of the group described above. The selected participants differed from the eliminated ones only in the accuracy score (at least 75% for the former and less than 75% for the latter). Interpreting students and non-interpreting trilinguals could be treated as a single group since no group effect was found in the analyses described in previous sections. However, in order to double check for any group effects in the new group a non-parametric Mann-Whitney test was performed. The fact that, the test yielded no significant result ($p > .05$) allowed for the selected participants to be treated as a homogenous sample.

Results and discussion

Data included in this analysis and other analyses reported on in the paragraphs to follow were trimmed as described above. In order to analyse data obtained from the redefined group of participants, a Repeated Measures ANOVA on participants' mean RTs was performed. Table 3 presents participants' mean RTs (in ms), SD (in parentheses) and error rates (ER, in %).

The Repeated Measures ANOVA yielded a significant main effect of the target language ($F(1,19) = 63,171$, $p < .05$) and a significant interaction between context and target language ($F(1,19) = 13,841$, $p < .05$). L3 nouns were translated 273 ms faster into L1 than into L2. Moreover, L3–L1 translation was 424 ms faster in the HC condition than in the LC condition and the L3–L2 translation was 175 ms faster in HC than in LC condition.

There were also main effects of context ($F(1,19) = 118.48$ $p < .05$) and of the cognate status ($F(1,19) = 1.911$ $p < .05$). However, the interaction between cognate status and context failed to reach the significance level ($F(1,19) = 0.356$ $p > .05$). This means that generally the speed of translation was affected by context and the cognate status. In order to investigate the influence of context on the processing of cognates depending on the target language a set of a priori planned simple effects analyses within the two levels of the factor context were performed. It needs to be noted that even though in the materials we used only L3–L2 cognates (no L3–L1 cognates were included), when analysing data, a t-test with two levels of cognate status was performed not only for the L3–L2

Table 3. Mean RTs (in ms), SDs (in parentheses) and error rates (in %) in each condition in the two translation directions.

		L3–L1 translation		L3–L2 translation	
Conditions		Mean (SD)	ER	Mean (SD)	ER
HC	Cognates	801 (172)	0.8	1002 (322)	3.3
	Non-cognates	896 (202)	3.5	1007 (279)	3.8
LC	Cognates	996 (175)	0.0	1430 (224)	0.8
	Non-cognates	1051 (170)	9.4	1424 (232)	5.1

translation but also for the L3–L1 translation in order to check for any possible influence of L3–L2 cognates on the production of L1 translation equivalents.

The a priori planned simple effect analyses revealed a significant cognate facilitation effect in both context constraints but only in the L3–L1 translation. These analyses involved paired samples t-tests with Bonferroni correction applied, resulting in a significance level set at $p < .025$. In HC cognates were translated 96 ms faster than non-cognates ($t(19) = -2.80$, $p < .025$), and in LC the cognate facilitation effect was 55 ms ($t(19) = -2.44$, $p < .025$). However, no such effect was found for the L3–L2 translation (in HC $t(19) = -.07$, $p > .025$; in LC $t(19) = .13$, $p > .025$). These results are particularly interesting in view of the fact that only L3–L2 cognates were employed in the study.

As it transpires from the data, translation from the third language (English) was significantly faster into the first language (Polish) than into the second language (German). This effect, according to Francis and Gallard (2005) may be ascribed to proficiency. The process of translation involves two subsequent processes – comprehension and production. Since production is more sensitive to proficiency effects than comprehension, the net effect favours production into the dominant language (L1) rather than into the weaker L2. This finding is also in line with Kujałowicz (2007).

The fact that context effects were found in translation lends further support to Van Hell (2005) and Van Hell and De Groot (2008). However, it needs to be noted that our data are not entirely in accord with the results from the two aforementioned studies. As has been described above, Van Hell (2005) as well as Van Hell and De Groot (2008) found significant cognate facilitation effects only in the LC condition which they interpreted as an indication of the fact that the semantically constraining context language limits non-selectivity in lexical processing. In the present study, however, cognate facilitation was recorded in both context constraints but only in L3–L1 translation. This observation is intriguing for three reasons.

Firstly, in our research we employed L3–L2 cognates (not L3–L1 ones) so they should have significantly affected processing in the L3–L2 translation more than in the L3–L1 one (see Tymczyńska, 2012). However counter-intuitive our data are, they still indicate language non-selectivity in lexical processing exhibited by the trilingual participants. If L3–L2 cognates influenced translation from L3 to L1, it means that L2 translation equivalents must have been pre-activated to some degree when orthographically similar L3 stimuli were presented.

Secondly, the absence of cognate facilitation effect in the L3–L2 translation can be a result of inhibition in the production of L3–L2 translations, especially that many of our participants reported difficulties with the production of L2 equivalents in response to L3 items in general. The fact that accuracy in L3–L2 translation fell below 75% for more than a half of the participants also suggests that translation to another foreign language was a rather difficult task. As has already been mentioned, Francis and Gallard (2005) suggested that translation is composed of two processes – comprehension and production, hence it might have been the case that cognate facilitation effect in comprehension was attenuated by inhibition in production.

Finally, the results of our study may be interpreted as a reflection of the effects of learning experience/learning strategies applied by our participants, which in turn influenced the quality of lexical connections in the participants' mental lexicons. The fact that we observed cognate facilitation induced by L3–L2 cognates in L3–L1 translation but not in L3–L2 translation has important implications for RHM (Kroll & Stewart, 1994) and for BIA+ or its multilingual extension – MIA (Dijkstra, 2003).

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However, since RHM was designed to model translation in bilinguals, a number of assumptions have to be made before RHM can be extended to our population of trilinguals.

Since, with all probability, most of our participants have acquired L3 in the context of L1 (Polish), which is typical in the Polish educational system, the majority of the participants are likely to have used L3–L1 translation and L1–L3 translation as a vocabulary learning strategy. By the same token, the participants are very unlikely to have used L3–L2 translation in their learning experience. Consequently, this might have induced strong L3–L1 lexical links and significantly weaker (or almost nonexistent) L3–L2 lexical links. Furthermore, our participants, being quite proficient in L3, must have already developed conceptual links between L3 and the conceptual store (see Francis & Gallard, 2005). Hence, as a result of learning experience, L3 words are strongly connected (via lexical links) to their L1 equivalents and weakly connected to their L2 equivalents. Simultaneously, L3 items are already linked to the conceptual store via the conceptual links. Such a makeup of connections in the mental lexicons of our participants can explain why we observed L3–L2 cognate facilitation in L3–L1 translation but not in L3–L2 translation. The mechanics of cognate facilitation (or its lack) will be presented in detail with the use of an example of an L3–L2 cognate word BUTTER.

When BUTTER is presented as a cue word in the L3–L1 translation task, it first needs to be recognised by the visual system. Following MIA (see Dijkstra, 2003; Dijkstra et al., 2010), we assume that lexical representations of BUTTER get activated in L3 as well as in L2 due to the overlap in orthography and semantics of the representations. Then, following RHM, it can be assumed that by way of lexical links between translation equivalents, this activation travels to the L1 equivalent (MASŁO). Consequently, MASŁO receives activation from two sources (from the L3 and the L2 representations of BUTTER) and these joint jolts of activation give MASŁO a head start in translation. In contrast, when the to-be-translated L3 word is a non-cognate, the activation received by its L1 equivalent via lexical links comes only from one source i.e. the L3 representation. Hence, L1 equivalents are produced faster in response to L2–L3 cognates than in response to L3 non-cognates.

In turn, the lack of cognate facilitation in L3–L2 translation can be explained as a consequence of the quality of connections between L3 and L2 equivalents. When the L3–L2 cognate (e.g. BUTTER) is presented as a cue word in the translation task, activation cannot travel between L2 and L3 lexical representations (as the links are weak), hence, the activation travels from L3 lexical representation to its conceptual representation and only then (via conceptual links) to the lexical representation of BUTTER in L2. The same takes place when L3 non-cognate words are translated into L2. Consequently, as a result of the weakness (or even lack) of lexical links between L2 and L3 items, the L3–L2 translation of cognates and non-cognates lasts equally long as in either case the translation is executed via the conceptual links, which attenuate the facilitative (lexical) effects of cognates.

Another interesting issue is the fact that cognate facilitation was not only present in HC but it was also almost twice the size of the cognate facilitation observed in the LC condition. This observation runs counter to other studies employing context, which have been reported to this date. This discrepancy might either be an artefact, a result of noise in the data due to a small sample, or it can result from the qualitatively different processing in trilingual speakers.

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General conclusions

In this study we investigated the influence of sentence context on cognate facilitation in L3–L2 and in L3–L1 translation production. We also aimed to compare interpreting trainees with non-interpreters to test whether interpreting training leads to boosted performance in the tested translation directions. The data obtained suggest that translation from L3 to L2 was much more difficult for both groups of participants (very low accuracy rate) than from L3 to L1. Hence the former translation direction had to be initially excluded from analysis and it could only be analysed once the design of the study was redefined.

In general, our results provided evidence for the effect of sentence context on translation production leading to shorter latencies in a semantically constraining context than in a non-constraining one. Also, we found significant cognate facilitation effect in L3–L1 translation, in that cognates were generally translated faster than non-cognates. This finding is especially interesting if we take into account the fact that the study employed only L3–L2 cognates and no L3–L1 ones. This pattern of results has implications for the representation of translation equivalents within RHM, and it seems to indicate that, as a result of learning experience, at the lexical level L3 words are strongly connected to their L1 equivalents but only weakly (or not at all) to the L2 ones. The observation of possible influence of participants' educational experience on the structure of their mental lexicons and lexical processing may prove useful in further research.

In turn, cognate facilitation effect was found to be modulated by sentence context only once the study design was redefined. However, the observed pattern of the modulation is at odds with similar studies with bilinguals. This might be a result of noise in the data or a peculiarity of lexical processing in L3. In order to address this problem in a more systematic manner, further research with trilingual speakers seems to be in place. Future experimentation should in particular address the question as to why cognate facilitation is not observed in L3–L2 translation, as well as to how sentence context influences language processing in trilinguals.

Another issue which calls for further research is the fact that we observed no significant differences in the use of sentence context by interpreting trainees and by non-interpreting trilinguals. This might be interpreted as evidence for the lack of accelerated semantic processing on the part of interpreting trainees, which would be in line with the results obtained by Christoffels et al. (2006), who found no difference in single word translation tasks between interpreters and non-interpreting English teachers and concluded that efficient word retrieval was not uniquely relevant for interpreting. Since the present study involved translation directions which had not been practised during interpreting training before the study, we could not conclude whether the lack of the difference between groups stemmed from the selected translation directions or from the insufficient amount of interpreting training, which had not yet led to observable differences in the speed of semantic processing in participants. Hence, further research should involve the comparison of translation performance in the practiced translation directions and also at different stages of interpreting training.

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